

ZORK II

The Wizard of Frobozz — A Concise 400-Point Walkthrough, with an Opcode Census

Infocom, 1981 (`stories/zork2.z3`, Z-machine version 3), played on the zAI-Machine interpreter with random seed 42. Final result: **400 of a possible 400, in 454 moves**, rank *Master Adventurer*, ending on the stairs down to Zork III. Every command below is replayed by `ZorkIIPerfectScore`, which also tallies each opcode executed (see the Opcode Census at the end).

Solved during play with the game's manual as the only outside reference (it is the generic Zork instruction booklet and gives no puzzle hints). No published walkthrough or hints were consulted. The interpreter's disassembler and object/global inspection tooling were used to understand puzzles.

Know Before You Go

- **The lamp is your clock.** It gives only about 270 turns of light. Rooms with glowing moss or other light (the garden, cavern, Tea Room, Workroom, Pentagram Room) let you turn it off. This route runs it close.
- **The Wizard is a nuisance, not a killer.** About 10% of turns he appears and casts an "F" spell. *Fence* blocks exits, *Feeble* cuts what you can carry, and *Fall* tips you out of baskets. Spells wear off. The script inserts WAITs at chosen points so his visits land where they do no harm.
- **The Carousel Room spins** until the robot pushes the triangular button, sending you out a random exit. Just walk back in and try again.
- **Treasures score when first taken.** Ten of them are later paid to the demon (+2 each), so stash them near the Wizard's workroom. There is no trophy case.
- **Inventory is weight-limited.** The route stashes things in the gazebo, the Pearl Room and the Carousel Room.
- **Death is survivable.** Up to a point: you are revived in the mist spheres near the barrow, but your possessions are scattered. This route never dies.

The Walkthrough

1. The Barrow to the Carousel (+0)

You start in the barrow with the lamp and an Elvish sword that glows near danger.

Command	What happens / why	Pts
VERBOSE, TAKE LAMP, TAKE SWORD, TURN ON LAMP, S, S, S, SW, S, SE, S, S, W	Through the Great Cavern and garden to the spinning Carousel Room.	
WAIT ×11, N	This random draw lands in Marble Hall.	

2. The Dragon (+20)

The dragon only follows while angry, so poke it once per room.

Command	What happens / why	Pts
N, N, W, N, HIT DRAGON WITH SWORD, S, HIT DRAGON WITH SWORD, S, HIT DRAGON WITH SWORD, W	In the Ice Room the dragon sees its reflection, breathes fire at the glacier, and drowns in the meltwater.	+5
W, TAKE RUBY, WAIT	The glacier's melting opens the Lava Room.	+15

3. The Princess (+35)

Command	What happens / why	Pts
E, E, N, N, N, OPEN CHEST, TAKE STATUETTE	The squeaky lid wakes the princess.	+20
S, E, WAIT, E, DOWN, S, E, E, N, WAIT ×4	Follow her (she opens the Marble Hall passage) to the garden. The unicorn arrives, and she gives you the gold key and a rose.	+15

4. The Tiny Room Door (+20)

The key is in the keyhole on the far side of a locked door.

Command	What happens / why	Pts
TURN OFF LAMP, IN, DROP RUBY, DROP STATUETTE, DROP ROSE, TAKE MAT, TAKE LETTER OPENER, TAKE MATCHBOOK, OUT, S, TURN ON LAMP, W, W, N, N, U	Stash in the gazebo.	
OPEN LID, PUT MAT UNDER DOOR, PUT LETTER OPENER IN KEYHOLE, PULL MAT, TAKE KEY, TAKE LETTER OPENER, UNLOCK DOOR WITH IRON KEY, OPEN DOOR, N, TAKE SPHERE	The key falls onto the mat.	+20

5. The Bank of Zork (+45)

The curtain teleports you to the room linked to the doorway you last entered by. Walking through that room's "thin" wall brings you back. Treasures can't leave through the doors (an alarm sounds), but they can leave through the curtain.

Command	What happens / why	Pts
S, D, W, N, W, W, DROP SWORD, DROP LETTER OPENER, DROP RUSTY KEY, DROP MATCHBOOK		
NE, E, S, TAKE PORTRAIT, N	Entering from the south aims the curtain at the Small Room.	+20
ENTER CURTAIN, WALK THROUGH SOUTH WALL, ENTER CURTAIN, WAIT, WAIT, TAKE BILLS	Small Room, then the Vault. The waits let a Feeble spell wear off.	+25
WALK THROUGH NORTH WALL, DROP BILLS, DROP PORTRAIT, E, E, TAKE BILLS, TAKE PORTRAIT, ENTER CURTAIN, S, TAKE SPHERE, TAKE MATCHBOOK	Re-enter from the east so the curtain leads to the East Viewing Room.	

6. The Riddle (+5)

Command	What happens / why	Pts
E, E, S, S, SE, SE, S, SE	Back to the carousel; two tries land in the Riddle Room.	
ANSWER "WELL", E, DROP PORTRAIT, DROP BILLS, W, NW		+5

7. Up the Well (+25)

"Water makes the bucket go."

Command	What happens / why	Pts
E, SW, E, S, E, N, N, IN, TAKE TEAPOT, TAKE NEWSPAPER, OUT, S, W, W, N, FILL TEAPOT WITH WATER, S, S		
SE, SE, SE, E, TAKE NECKLACE, E, GET IN BUCKET, POUR WATER IN BUCKET, GET OUT OF BUCKET		+25

8. The Mad Tea Party (+15)

Command	What happens / why	Pts
E, DROP TEAPOT, DROP NEWSPAPER, TAKE GREEN CAKE, TAKE RED CAKE, TAKE BLUE CAKE, TAKE ORANGE CAKE, TURN OFF LAMP, WAIT ×7	Wait out a Fall spell.	
EAT GREEN CAKE, TURN ON LAMP, E, THROW RED CAKE IN POOL, TAKE CANDY, TAKE FLASK, W, EAT BLUE CAKE, TURN OFF LAMP	"Eat Me" shrinks you through the hole. The red cake evaporates the pool of tears, and the blue cake restores you.	+15

9. The Robot (+20)

Command	What happens / why	Pts
NW, TAKE PAPER, ROBOT, GO EAST, E, ROBOT, GO SOUTH, S, TAKE SPHERE, ROBOT, LIFT CAGE, DROP PAPER, TAKE SPHERE	A cage traps you with poison gas; the robot hurls it away.	+20
ROBOT, GO NORTH, N, ROBOT, PUSH TRIANGULAR BUTTON	Stops the carousel (and sets the Low Room spinning instead).	
W, SE, W, SE, DROP ORANGE CAKE, TAKE TEAPOT, TURN ON LAMP, W, GET IN BUCKET, FILL TEAPOT WITH WATER, GET OUT OF BUCKET, W, W	Take the water out of the bucket to go back down.	

10. The Wizard's Domain (+50)

Command	What happens / why	Pts
NW, OPEN BOX, DROP TEAPOT, TAKE VIOLIN	The steel box appeared when the carousel stopped.	+20
SW, SW, GIVE CANDY TO LIZARD, UNLOCK DOOR WITH GOLD KEY, OPEN DOOR, S		+10
W, PUT RED SPHERE ON RUBY STAND, PUT BLUE SPHERE ON SAPPHIRE STAND, DROP NECKLACE, DROP VIOLIN, DROP FLASK		
E, N, N, NE, TAKE BOX, SW, SW, S, W, W, THROW BOX AT AQUARIUM, TAKE SPHERE, E, PUT SPHERE ON DIAMOND STAND	The serpent suffocates. The three spheres fuse into a black sphere.	+20

11. Supplies for the Volcano (+0)

Command	What happens / why	Pts
TURN OFF LAMP, WAIT ×3, TURN ON LAMP	Timing, so the Wizard's Fall spell misses the balloon launch.	
E, N, N, TAKE STRING, NE, N, TAKE BRICK, N, N, W, N, W, W, TAKE MATCHBOOK, NE, E, TAKE BROCHURE, E, S, E, E, S, S, W, W, S	Fuse, explosive, matches and fuel, then down to the Volcano Bottom.	

12. The Balloon (+50)

Burning fuel lifts the balloon one level per turn. CLOSE RECEPTACLE lets it sink. LAND, then TIE WIRE TO HOOK.

Command	What happens / why	Pts
GET IN BASKET, OPEN RECEPTACLE, PUT BROCHURE IN RECEPTACLE, LIGHT MATCH, LIGHT BROCHURE WITH MATCH, WAIT ×4, LAND, TIE WIRE TO HOOK, GET OUT OF BASKET, S		
PUT BRICK IN HOLE, PUT STRING IN BRICK, LIGHT MATCH, LIGHT STRING WITH MATCH, N, S, TAKE CROWN, TAKE CARD, N	<i>Dash back in after the blast, before the room collapses.</i>	+20
GET IN BASKET, CLOSE RECEPTACLE, UNTIE WIRE, WAIT ×4, LAND, TIE WIRE TO HOOK, GET OUT OF BASKET, TAKE COIN	<i>Leave before the Wide Ledge collapses.</i>	+20
S, TAKE PURPLE BOOK, OPEN PURPLE BOOK, TAKE STAMP, N, GET IN BASKET, UNTIE WIRE, WAIT ×3	<i>The balloon lands at the bottom and breaks.</i>	+10

13. The Demon (+80)

Gather ten treasures and summon the demon from the black sphere.

Command	What happens / why	Pts
N, DROP PURPLE BOOK, DROP CARD, E, E, SE, SE, E, TAKE BILLS, TAKE PORTRAIT, W, NW, E, N, N, TURN OFF LAMP, IN, TAKE RUBY, TAKE STATUETTE, OUT, S, S, TURN ON LAMP, W, SW, SW, S, W		
TURN OFF LAMP, DROP LAMP, DROP MATCHBOOK, TAKE BLACK SPHERE		+30
S, PUT BLACK SPHERE IN CIRCLE, GIVE CROWN / COIN / STAMP / BILLS / PORTRAIT / RUBY / STATUETTE / GOLD KEY TO DEMON, N, TAKE VIOLIN, TAKE NECKLACE, S, GIVE VIOLIN / NECKLACE TO DEMON	<i>Each gift is worth 2 points. The tenth pays his fee.</i>	+20
DEMON, GIVE ME THE WAND, TAKE WAND		+30

14. The Menhir (+15)

Command	What happens / why	Pts
N, TAKE LAMP, TURN ON LAMP, E, N, N, NE, S, POINT WAND AT MENHIR, SAY "FLOAT", SW, TAKE COLLAR		+15

15. Cerberus and the Crypt (+20)

Command	What happens / why	Pts
NE, S, D, D, SE, NE, NW, SW	<i>In the oddly-angled rooms, this sequence swings open the floor at the foot of the stairway.</i>	+5
E, D, PUT COLLAR ON DOG	<i>Cerberus becomes a puppy.</i>	+10
E, OPEN DOOR, S	<i>"Feel Free."</i>	+5
CLOSE DOOR, TURN OFF LAMP, OPEN SECRET DOOR, S	<i>In the dark, an "F" outlines the way on to Zork III.</i>	

Where the Points Come From

Event	Pts	Event	Pts
Dragon drowns	5	Wizard's Workshop	10
Ruby	15	Clear crystal sphere	20
Golden dragon statuette	20	Gaudy crown	20
Princess's gift	15	Priceless zorkmid	20
Blue crystal sphere	20	Flathead stamp	10
Portrait of Flathead	20	Black crystal sphere	30
Zorkmid bills	25	Ten gifts to the demon	20
Riddle answered	5	Wizard's wand	30
Pearl necklace	15	Dog collar	15
Top of Well	10	Oddly-angled maze	5
Candy	15	Cerberus tamed	10
Red crystal sphere	20	Crypt anteroom	3
Fancy violin	20	Crypt	2

Opcode Census

Every instruction executed during the 454-move run was decoded with the interpreter's own disassembler and counted by mnemonic. The run used **52 distinct opcodes** over **372,688** instructions. As a version 3 story, Zork II uses the classic call (`call_vs` here), `read` (`sread`) and the stack opcodes `push/pull`. It ends with `quit` after the victory text.

#	Opcode	Form:No.	Executions	Purpose
1	<code>jz</code>	1OP:128	72,435	branch if zero
2	<code>je</code>	2OP:1	50,730	branch if equal to any operand
3	<code>call_vs</code>	VAR:224	26,257	call routine (V3 "call"), store result
4	<code>store</code>	2OP:13	25,471	store to variable
5	<code>test_attr</code>	2OP:10	23,656	branch if object has attribute
6	<code>loadw</code>	2OP:15	22,548	load word from array
7	<code>jump</code>	1OP:140	20,931	unconditional jump
8	<code>add</code>	2OP:20	14,968	signed addition
9	<code>storew</code>	VAR:225	10,588	store word into array
10	<code>inc_chk</code>	2OP:5	9,247	increment and branch if greater
11	<code>get_prop_addr</code>	2OP:18	8,946	address of property data
12	<code>get_sibling</code>	1OP:129	8,431	next sibling, branch if any
13	<code>loadb</code>	2OP:16	8,227	load byte from array
14	<code>get_child</code>	1OP:130	8,189	first child object, branch if any
15	<code>jg</code>	2OP:3	6,190	branch if greater (signed)
16	<code>rfalse</code>	0OP:177	5,778	return false
17	<code>test</code>	2OP:7	5,060	branch if all bitmap flags set
18	<code>rtrue</code>	0OP:176	4,004	return true
19	<code>get_parent</code>	1OP:131	3,776	parent object
20	<code>sub</code>	2OP:21	3,773	signed subtraction
21	<code>jl</code>	2OP:2	3,679	branch if less (signed)
22	<code>get_prop</code>	2OP:17	3,481	read property value
23	<code>ret</code>	1OP:139	3,210	return value
24	<code>ret_popped</code>	0OP:184	3,107	return top of stack
25	<code>push</code>	VAR:232	2,882	push onto stack
26	<code>get_prop_len</code>	1OP:132	2,539	length of property data
27	<code>dec_chk</code>	2OP:4	1,969	decrement and branch if less

#	Opcode	Form:No.	Executions	Purpose
28	div	2OP:23	1,807	signed division
29	and	2OP:9	1,624	bitwise AND
30	new_line	0OP:187	1,225	print newline
31	print	0OP:178	995	print inline string
32	jin	2OP:6	982	branch if object is a child of
33	inc	1OP:133	912	increment variable
34	mul	2OP:22	833	signed multiplication
35	get_next_prop	2OP:19	733	next property number of an object
36	storeb	VAR:226	418	store byte into array
37	pull	VAR:233	415	pop stack into variable
38	set_attr	2OP:11	403	set object attribute
39	read	VAR:228	391	read a line of input (V3 sread)
40	print_obj	1OP:138	373	print object short name
41	insert_obj	2OP:14	355	move object into another
42	clear_attr	2OP:12	340	clear object attribute
43	print_paddr	1OP:141	340	print string at packed address
44	print_ret	0OP:179	189	print inline string, newline, return true
45	random	VAR:231	168	random number / reseed
46	put_prop	VAR:227	45	write property value
47	remove_obj	1OP:137	45	detach object from parent
48	load	1OP:142	11	load variable value (no pop)
49	print_char	VAR:229	6	print ZSCII character
50	print_num	VAR:230	3	print signed number
51	dec	1OP:134	2	decrement variable
52	quit	0OP:186	1	end the game

Full Command List

One line per input, exactly as replayed by the test (seed 42). Read down each column.

1 verbose	99 north	197 west	295 wait
2 take lamp	100 enter curtain	198 southeast	296 wait
3 take sword	101 walk through south wall	199 drop orange cake	297 wait
4 turn on lamp	102 enter curtain	200 take teapot	298 land
5 south	103 wait	201 turn on lamp	299 tie wire to hook
6 south	104 wait	202 west	300 get out of basket
7 south	105 take bills	203 get in bucket	301 take coin
8 southwest	106 walk through north wall	204 fill teapot with water	302 south
9 south	107 drop bills	205 get out of bucket	303 take purple book
10 southeast	108 drop portrait	206 west	304 open purple book
11 south	109 east	207 west	305 take stamp
12 south	110 east	208 northwest	306 north
13 west	111 take bills	209 open box	307 get in basket
14 wait	112 take portrait	210 drop teapot	308 untie wire
15 wait	113 enter curtain	211 take violin	309 wait
16 wait	114 south	212 southwest	310 wait
17 wait	115 take sphere	213 southwest	311 wait
18 wait	116 take matchbook	214 give candy to lizard	312 north
19 wait	117 east	215 unlock door with gold key	313 drop purple book
20 wait	118 east	216 open door	314 drop card
21 wait	119 south	217 south	315 east
22 wait	120 south	218 west	316 east
23 wait	121 southeast	219 put red sphere on ruby st	317 southeast
24 wait	122 southeast	220 put blue sphere on sapphi	318 southeast
25 north	123 south	221 drop necklace	319 east
26 north	124 southeast	222 drop violin	320 take bills
27 north	125 answer "well"	223 drop flask	321 take portrait
28 west	126 east	224 east	322 west
29 north	127 drop portrait	225 north	323 northwest
30 hit dragon with sword	128 drop bills	226 north	324 east
31 south	129 west	227 northeast	325 north
32 hit dragon with sword	130 northwest	228 take box	326 north
33 south	131 east	229 southwest	327 turn off lamp
34 hit dragon with sword	132 southwest	230 southwest	328 in
35 west	133 east	231 south	329 take ruby
36 west	134 south	232 west	330 take statuette
37 take ruby	135 east	233 west	331 out
38 wait	136 north	234 throw box at aquarium	332 south
39 east	137 north	235 take sphere	333 south
40 east	138 in	236 east	334 turn on lamp
41 north	139 take teapot	237 put sphere on diamond sta	335 west
42 north	140 take newspaper	238 turn off lamp	336 southwest
43 north	141 out	239 wait	337 southwest
44 open chest	142 south	240 wait	338 south
45 take statuette	143 west	241 wait	339 west
46 south	144 west	242 turn on lamp	340 turn off lamp
47 east	145 north	243 east	341 drop lamp
48 wait	146 fill teapot with water	244 north	342 drop matchbook
49 east	147 south	245 north	343 take black sphere
50 down	148 south	246 take string	344 south
51 south	149 southeast	247 northeast	345 put black sphere in circl
52 east	150 southeast	248 north	346 give crown to demon
53 east	151 southeast	249 take brick	347 give coin to demon
54 north	152 east	250 north	348 give stamp to demon
55 wait	153 take necklace	251 north	349 give bills to demon
56 wait	154 east	252 west	350 give portrait to demon
57 wait	155 get in bucket	253 north	351 give ruby to demon
58 wait	156 pour water in bucket	254 west	352 give statuette to demon
59 turn off lamp	157 get out of bucket	255 west	353 give gold key to demon
60 in	158 east	256 take matchbook	354 north
61 drop ruby	159 drop teapot	257 northeast	355 take violin
62 drop statuette	160 drop newspaper	258 east	356 take necklace
63 drop rose	161 take green cake	259 take brochure	357 south
64 take mat	162 take red cake	260 east	358 give violin to demon
65 take letter opener	163 take blue cake	261 south	359 give necklace to demon
66 take matchbook	164 take orange cake	262 east	360 demon, give me the wand
67 out	165 turn off lamp	263 east	361 take wand
68 south	166 wait	264 south	362 north
69 turn on lamp	167 wait	265 south	363 take lamp
70 west	168 wait	266 west	364 turn on lamp
71 west	169 wait	267 west	365 east
72 north	170 wait	268 south	366 north
73 north	171 wait	269 get in basket	367 north
74 up	172 wait	270 open receptacle	368 northeast
75 open lid	173 eat green cake	271 put brochure in receptacl	369 south
76 put mat under door	174 turn on lamp	272 light match	370 point wand at menhir
77 put letter opener in keyh	175 east	273 light brochure with match	371 say "float"
78 pull mat	176 throw red cake in pool	274 wait	372 southwest
79 take key	177 take candy	275 wait	373 take collar
80 take letter opener	178 take flask	276 wait	374 northeast
81 unlock door with iron key	179 west	277 wait	375 south
82 open door	180 eat blue cake	278 land	376 down
83 north	181 turn off lamp	279 tie wire to hook	377 down
84 take sphere	182 northwest	280 get out of basket	378 southeast
85 south	183 take paper	281 south	379 northeast
86 down	184 robot, go east	282 put brick in hole	380 northwest

87 west	185 east	283 put string in brick	381 southwest
88 north	186 robot, go south	284 light match	382 east
89 west	187 south	285 light string with match	383 down
90 west	188 take sphere	286 north	384 put collar on dog
91 drop sword	189 robot, lift cage	287 south	385 east
92 drop letter opener	190 drop paper	288 take crown	386 open door
93 drop rusty key	191 take sphere	289 take card	387 south
94 drop matchbook	192 robot, go north	290 north	388 close door
95 northeast	193 north	291 get in basket	389 turn off lamp
96 east	194 robot, push triangular bu	292 close receptacle	390 open secret door
97 south	195 west	293 untie wire	391 south
98 take portrait	196 southeast	294 wait	